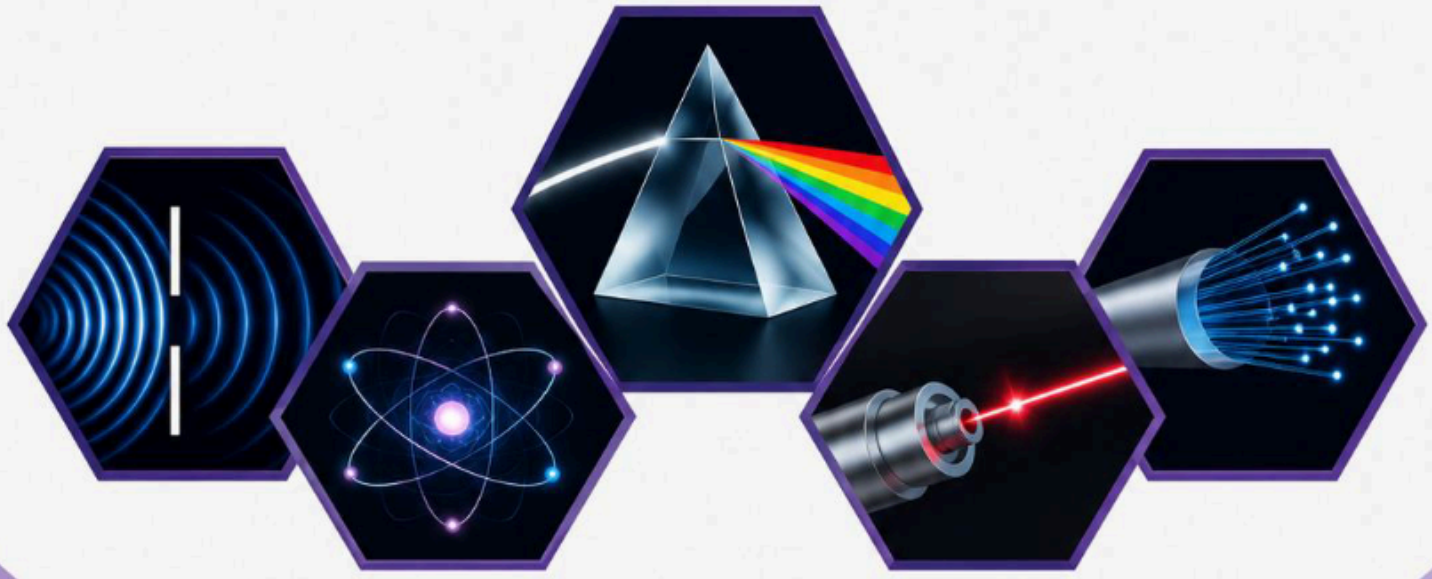
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ENGINEERING PHYSICS-I

PART 1

QUANTUM MECHANICS | WAVE OPTICS | LASERS | OPTICAL FIBER



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QUANTUM MECHANICS

Quantum mechanics is the branch of physics that describes the behavior of matter and energy at microscopic scale (atomic and subatomic level) where classical laws of physics fail.

1. NEED FOR QUANTUM MECHANICS

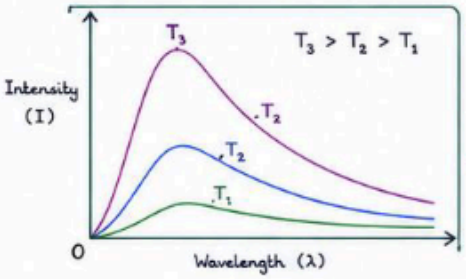
- Classical physics could not explain the behavior of radiation emitted by hot bodies.
- Photoelectric effect could not be explained by wave theory of light.
- Stability of atoms and discrete line spectra of elements remained unexplained.
- These failures led to the development of quantum theory.

CLASSICAL vs QUANTUM WORLD

Classical World	Quantum World
• Continuous energy	• Discrete energy
• Deterministic	• Probabilistic
• Exact path known	• Uncertain path
• Macroscopic scale	• Microscopic scale

2. BLACK-BODY RADIATION

A black body is an ideal body which absorbs all incident radiation and does not reflect any. It emits maximum possible radiation at all wavelengths for a given temperature.



★ As temperature increases, peak shifts to smaller wavelength.

★ Area under curve increases with temperature.

3. PLANCK'S QUANTUM THEORY

Max Planck (1900) proposed that energy is emitted or absorbed in discrete packets called quanta.

Energy of each quantum,

$$E = h\nu$$

where,
h = Planck's constant = 6.626×10^{-34} J·s
ν = frequency of radiation

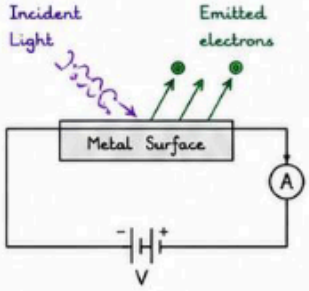
Key Points :

★ Energy is not continuous but quantized.

★ Higher the frequency, higher is the energy.

4. PHOTOELECTRIC EFFECT

Emission of electrons from a metal surface when light of suitable frequency is incident on it.



Observations

1. No time lag.
2. Maximum kinetic energy depends on frequency, not intensity.
3. There exists a threshold frequency (ν_0) below which no electrons are emitted.

5. EINSTEIN'S PHOTOELECTRIC EQUATION

Einstein explained the photoelectric effect by assuming light consists of quanta (photons).

$$h\nu = \phi + K_{max}$$

where,
$h\nu$ = energy of incident photon
ϕ = work function (minimum energy to liberate electron)
K_{max} = maximum kinetic energy of emitted electron

Important Relations

$$\nu_0 = \frac{\phi}{h} \quad \text{(threshold frequency)}$$

$$K_{max} = h(\nu - \nu_0)$$

★ Quantum theory introduced the idea that nature behaves in discrete packets of energy, laying the foundation of modern physics.

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SCHRÖDINGER WAVE EQUATIONS

The quantum state of a particle is completely described by a function ψ (psi) called wave function. Schrödinger developed a wave equation which governs the behavior of microscopic particles.

1. WAVE FUNCTION (ψ)

- $\psi(x,t)$ represents the quantum state of a particle.
- ψ is generally a complex function.
- $|\psi|^2$ gives the probability density of finding the particle at a point.
- $|\psi|^2 \, d\tau$ is the probability of finding the particle within volume $d\tau$.
- Total probability in all space = 1

$$\int_{\text{all space}} |\psi|^2 \, d\tau = 1$$

2. TIME-DEPENDENT SCHRÖDINGER EQUATION

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(x,t) + V(x,t)\psi(x,t)$$

where,

$\psi(x,t)$ = wave function

$i = \sqrt{-1}$

$\hbar = \frac{h}{2\pi}$ (reduced Planck's constant)

m = mass of particle

$V(x,t)$ = potential energy

∇^2 = Laplacian operator

In one dimension

$$\nabla^2 = \frac{\partial^2}{\partial x^2}$$

So,

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x,t)\psi$$

3. TIME-INDEPENDENT SCHRÖDINGER EQUATION

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(x) + V(x)\psi(x) = E\psi(x)$$

where,

$\psi(x)$ = wave function (depends only on position)

$V(x)$ = potential energy (only position dependent)

E = total energy of the particle (constant)

4. RELATION BETWEEN $\psi(x,t)$ AND $\psi(x)$

When V is independent of time, the solution of time-dependent equation can be separated as,

$$\psi(x,t) = \psi(x)\, T(t)$$

Substituting and simplifying,

$$T(t) = e^{-iEt/\hbar} \quad \text{(time part)}$$

Hence,

$$\psi(x,t) = \psi(x)\, e^{-iEt/\hbar} \quad \text{(complete solution)}$$

5. PHYSICAL SIGNIFICANCE OF SCHRÖDINGER EQUATION

- Gives the allowed energy values (E) and corresponding wave functions (ψ).
- Applicable to systems where classical mechanics fails (atomic and subatomic scale).
- Forms the basis of quantum mechanics.
- Helps in calculating measurable quantities like probability, expectation values, etc.

6. PROBABILITY INTERPRETATION

- Probability density at a point = $|\psi|^2$
- Probability of finding particle between x_1 and x_2 (in 1D) = $\int_{x_1}^{x_2} |\psi(x)|^2 \, dx$
- Similarly in 3D = $\int_{\tau}^{\infty} |\psi(x,y,z)|^2 \, d\tau$

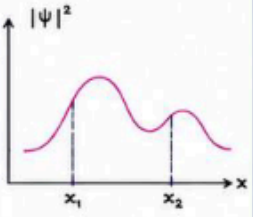
7. NORMALIZATION CONDITION

The total probability of finding the particle in all space is 1.

$$\int_{\text{all space}} |\psi|^2 \, d\tau = 1$$

In 1D $\rightarrow$ $\int_{-\infty}^{+\infty} |\psi(x)|^2 \, dx = 1$

In 3D $\rightarrow$ $\int_{\text{all space}}^{\infty} |\psi(x,y,z)|^2 \, d\tau = 1$



Probability = Area under curve between x_1 and x_2

8. IMPORTANT NOTES

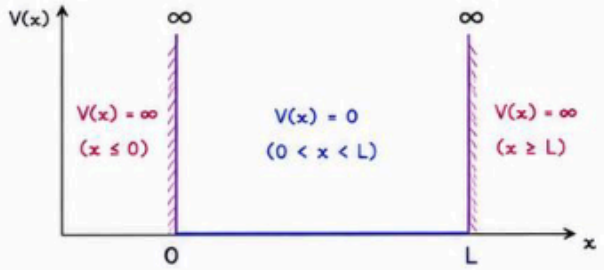
- ψ has no direct physical meaning; only $|\psi|^2$ has physical meaning.
- Schrödinger equation reduces to classical mechanics for large systems.
- Boundary conditions on ψ decide the allowed energy values.

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PARTICLE IN A 1D BOX & QUANTUM APPLICATIONS

A simple model where a particle is confined to move in a region of finite length with infinite potential barriers at the boundaries.

1. INFINITE POTENTIAL WELL (1D BOX)



- ★ The particle is confined in the region $0 < x < L$.
- ★ Outside this region, potential is infinite.
- ★ Wave function $\psi(x) = 0$ for $x \leq 0$ and $x \geq L$.
- ★ Inside the box $(0 < x < L)$, the particle is free.

2. TIME-INDEPENDENT SCHRÖDINGER EQUATION INSIDE THE BOX (0 < x < L)

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi(x)$$

General solution:

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

Using boundary conditions:

$\psi(0) = 0 \Rightarrow B = 0$

$\psi(L) = 0 \Rightarrow \sin(kL) = 0 \Rightarrow kL = n\pi$

$n = 1, 2, 3, \ldots$

Hence, $k_n = \frac{n\pi}{L}$ $(n = 1, 2, 3, \ldots)$

3. ENERGY EIGENVALUES

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2$$

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} \quad (n = 1, 2, 3, \ldots)$$

Energy levels are quantized and increase as n^2.

4. NORMALIZED WAVE FUNCTIONS

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right) \text{ for } 0 < x < L$$

$$0 \text{ elsewhere}$$

These wave functions are orthonormal:

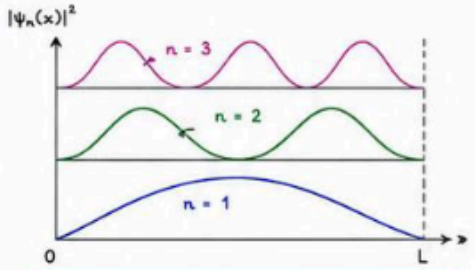
$$\int_0^L \psi_m(x)\psi_n(x)\,dx = \delta_{mn}$$

$\delta_{mn} = 1 \quad (m = n)$

$0 \quad (m \neq n)$

5. PROBABILITY DENSITY

$$|\psi_n(x)|^2 = \frac{2}{L}\sin^2\left(\frac{n\pi x}{L}\right)$$



Number of antinodes = n

Number of nodes (excluding boundaries) = n − 1

6. ZERO-POINT ENERGY

Lowest energy $(n = 1)$:

$$E_1 = \frac{\pi^2\hbar^2}{2mL^2} > 0$$

Even in the ground state, particle has non-zero energy. This is known as zero-point energy.

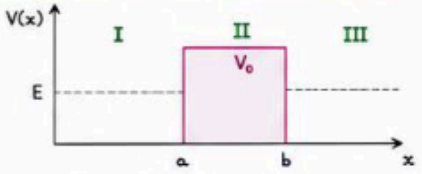
7. PROPERTIES

- Energy levels are discrete.
- $E_n \propto n^2$ (non-uniform spacing).
- Wave function is zero at the boundaries.
- Higher n ⇒ more oscillations.
- As L ↑, energy levels ↓.

8. QUANTUM APPLICATIONS

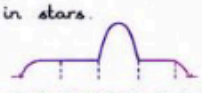
1. Quantum well structures in semiconductor devices.
2. Electrons in a metal cluster (confined in small regions).
3. Explains quantization of energy in nanoscale systems.
4. Basis for understanding molecular orbitals and bonding.
5. Models the behavior of particles in nanowires, quantum dots and thin films.

9. QUANTUM TUNNELLING (BRIEF)



A particle with energy $E < V_0$ has a finite probability of crossing the barrier and appearing in region III. This is a purely quantum phenomenon with no classical counterpart.

Applications: Tunnel diode, STM (Scanning Tunneling Microscope), Nuclear fusion in stars.



★ The 1D box problem is the foundation of quantum mechanics and helps in understanding how confinement leads to quantized energy and wave behavior.

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WAVE OPTICS — INTERFERENCE

Wave optics deals with the wave nature of light and phenomena that occur due to superposition of coherent light waves.

1. WAVE NATURE OF LIGHT

- ★ Light exhibits properties of both wave and particle.
- ★ Phenomena like interference, diffraction and polarization can only be explained on the basis of wave nature.
- ★ According to Huygens' principle, every point on a wavefront acts as a source of secondary wavelets.

2. SUPERPOSITION PRINCIPLE

When two or more coherent waves meet at a point, the resultant displacement is the algebraic sum of the individual displacements.

$$y = y_1 + y_2$$

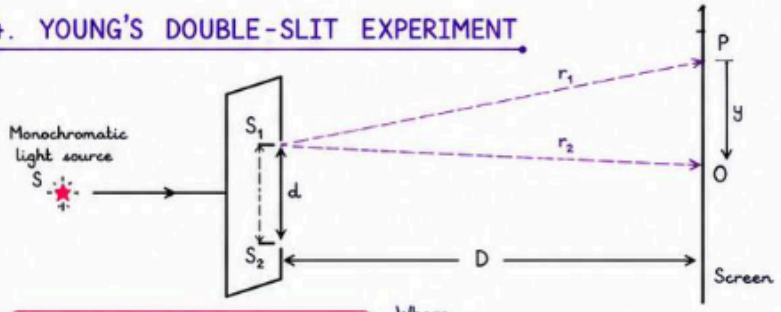
Constructive interference → waves add up

Destructive interference → waves cancel out

3. CONDITIONS FOR INTERFERENCE

1. The sources must be coherent (same frequency and constant phase difference).
2. The sources must have the same state of polarization.
3. The path difference between the waves should remain constant.

4. YOUNG'S DOUBLE-SLIT EXPERIMENT



Path difference,

$$\delta = r_2 - r_1 \approx d \sin\theta \approx d\left(\frac{y}{D}\right)$$

Where,

d = distance between slits

D = distance of screen from slits

y = distance of point P from O

5. CONDITIONS FOR MAXIMA & MINIMA

Bright Fringes (Constructive Interference)

Path difference, $\delta = n\lambda \quad (n = 0, \pm1, \pm2, \ldots)$

$$y_n = \frac{n\lambda D}{d}$$

n = order of maxima

Dark Fringes (Destructive Interference)

Path difference, $\delta = (2n+1)\frac{\lambda}{2} \quad (n = 0, \pm1, \pm2, \ldots)$

$$y_n = \frac{(2n+1)\lambda D}{2d}$$

n = order of minima

6. FRINGE WIDTH (β)

Fringe width is the distance between two consecutive bright (or dark) fringes.

$$\beta = \frac{\lambda D}{d}$$

It is constant for a given setup.

- ★ Independent of n (order).
- ★ Increases with λ and D.
- ★ Decreases with d.

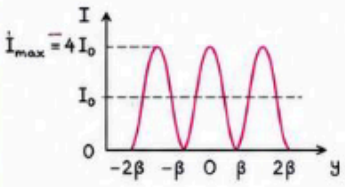
7. INTENSITY DISTRIBUTION

Intensity at point P,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$$

For equal intensities ($I_1 = I_2 = I_0$),

$$I = 4I_0\cos^2\left(\frac{\delta}{2}\right)$$



Bright fringe : $I = I_{max}$

Dark fringe : $I = 0$

8. KEY POINTS

- ★ Interference is the result of superposition of coherent waves.
- ★ Bright fringes are equally spaced.
- ★ Central fringe (n = 0) is always bright.
- ★ Fringe pattern is symmetric about the central maximum.
- ★ Experiment proves the wave nature of light.

★ Interference of light demonstrates that light behaves as a wave and the pattern depends on the path difference between the waves.



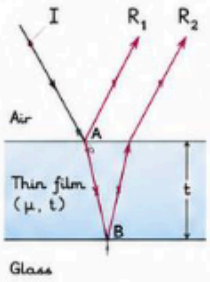
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THIN FILMS & NEWTON'S RINGS

When light is reflected from the two surfaces of a thin transparent film, the two reflected rays interfere with each other producing bright and dark fringes.

1. THIN FILMS – INTERFERENCE

- ★ A thin film is a transparent layer of thickness t and refractive index μ (μ_{film}) surrounded by a medium of refractive index μ_0.
- ★ Light reflected from top and bottom surfaces of the film leads to interference.
- ★ One of the reflected rays undergoes a phase change of π (or path change of $\frac{\lambda}{2}$) on reflection from a denser medium.
- ★ The other ray has no phase change.



2. CONDITIONS FOR INTERFERENCE

(A) Reflected Light

Path difference, $\delta = 2\mu t \cos r$

(one phase change of π)

(B) Transmitted Light

Path difference, $\delta = 2\mu t \cos r$

(no phase change)

} r = angle of refraction in the film

Reflected Light

Bright (Constructive):

$2\mu t \cos r = (2n+1)\frac{\lambda}{2}$

$(n = 0, 1, 2, \ldots)$

Dark (Destructive):

$2\mu t \cos r = n\lambda$

$(n = 0, 1, 2, \ldots)$

Transmitted Light

Bright (Constructive):

$2\mu t \cos r = n\lambda$

$(n = 0, 1, 2, \ldots)$

Dark (Destructive):

$2\mu t \cos r = (2n+1)\frac{\lambda}{2}$

$(n = 0, 1, 2, \ldots)$

3. SPECIAL CASE – NORMAL INCIDENCE (r = 0)

Reflected Light

Bright: $2\mu t = (2n+1)\frac{\lambda}{2}$

Dark: $2\mu t = n\lambda$

Transmitted Light

Bright: $2\mu t = n\lambda$

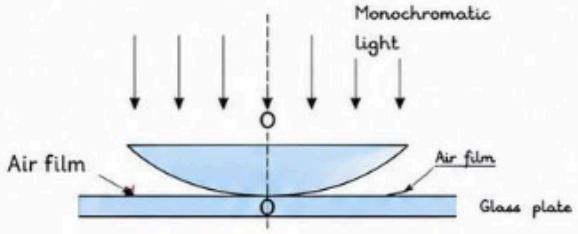
Dark: $2\mu t = (2n+1)\frac{\lambda}{2}$

4. IMPORTANT POINTS

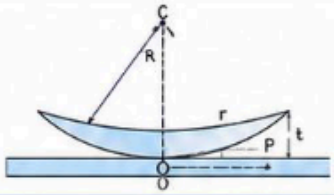
- ★ Fringes in thin films are mainly observed in reflected light.
- ★ Colors of thin films are due to interference.
- ★ Used in anti-reflection coatings, oil films on water, soap bubbles, etc.

5. NEWTON'S RINGS – BASIC IDEA

- ★ When a plano-convex lens is placed on a plane glass plate, an air film of varying thickness is formed between them.
- ★ Monochromatic light reflected from the top and bottom surfaces of this air film produces concentric bright and dark rings called Newton's rings.



6. THEORY AND FORMULAS



For a ring of radius r, thickness of air film,

$$t = \frac{r^2}{2R}$$

R = radius of curvature of the lens

Reflected Light (one phase change)

Bright rings:

$2t = (2n+1)\frac{\lambda}{2}$

$\Rightarrow 2\left(\frac{r^2}{2R}\right) = (2n+1)\frac{\lambda}{2}$

$\Rightarrow r_n^2 = (2n+1)\frac{\lambda R}{2}$

$(n = 0, 1, 2, \ldots)$

Dark rings:

$2t = n\lambda$

$\Rightarrow r_n^2 = n\lambda R$

$(n = 0, 1, 2, \ldots)$

Transmitted Light (no phase change)

Bright rings:

$2t = n\lambda$

$\Rightarrow r_n^2 = n\lambda R$

$(n = 0, 1, 2, \ldots)$

Dark rings:

$2t = (2n+1)\frac{\lambda}{2}$

$\Rightarrow r_n^2 = (2n+1)\frac{\lambda R}{2}$

$(n = 0, 1, 2, \ldots)$

7. KEY POINTS

- ★ Central spot is dark in reflected light ($n = 0$ gives $r_0 = 0$).
- ★ Central spot is bright in transmitted light.
- ★ Rings are equally spaced in r^2 (not in r).
- ★ Used to determine wavelength of monochromatic light.



Newton's rings

★ Thin films and Newton's rings beautifully demonstrate the wave nature of light through interference.



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DIFFRACTION OF LIGHT

Diffraction is the bending and spreading of light around the edges of an obstacle or through an aperture comparable to the wavelength of light.

1. INTRODUCTION

- ★ Diffraction is a wave phenomenon.
- ★ It occurs when light passes through a narrow slit or around the edges of an obstacle.
- ★ It is significant when the size of the aperture or obstacle is of the order of wavelength (λ).
- ★ It cannot be explained by geometrical optics.

2. TYPES OF DIFFRACTION

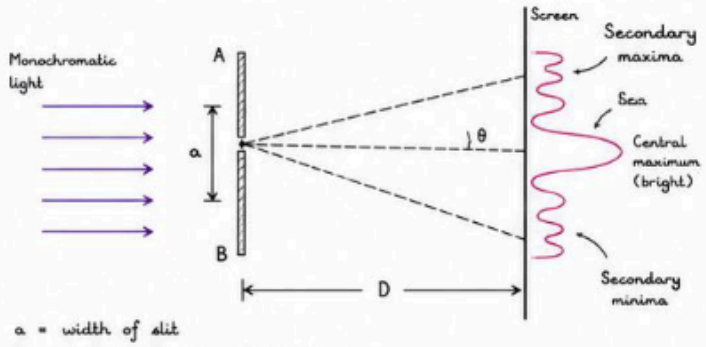
(A) Fresnel Diffraction

- Occurs when source or screen is at a finite distance.
- Wavefronts are curved.
- Pattern is complicated.

(B) Fraunhofer Diffraction

- Occurs when source and screen are at infinity.
- Wavefronts are plane and parallel.
- Pattern is simple and well-defined.

3. FRAUNHOFER DIFFRACTION DUE TO A SINGLE SLIT



a = width of slit
D = distance of screen from slit
θ = angle of diffraction

(A) Condition for Minima (Dark Fringes)

$$a \sin\theta = n\lambda \qquad (n = \pm 1, \pm 2, \pm 3, \ldots)$$

where,
a = width of slit
λ = wavelength of light
n = order of minima

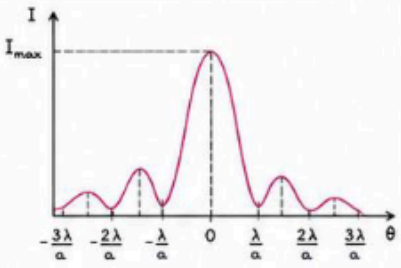
(B) Condition for Maxima (Bright Fringes)

$$a \sin\theta = \left(n + \frac{1}{2}\right)\lambda \qquad (n = 0, \pm 1, \pm 2, \ldots)$$

Important Point

★ Central maximum (n = 0) is always bright and has twice the width of other bright fringes.

4. INTENSITY DISTRIBUTION



Intensity at an angle θ is given by,

$$I = I_{max}\left(\frac{\sin\beta}{\beta}\right)^2$$

where,

$$\beta = \frac{\pi a \sin\theta}{\lambda}$$

5. ANGULAR WIDTH OF CENTRAL MAXIMUM

First minima on either side occur at,

$$\sin\theta = \frac{\lambda}{a}$$

$$\Rightarrow \text{Angular width } (2\theta) = 2\sin^{-1}\left(\frac{\lambda}{a}\right)$$

For small angles ($\sin\theta \approx \tan\theta \approx \theta$ in radians)

$$2\theta \approx 2\frac{\lambda}{a} \text{ (radians)}$$

★ Width of central maximum increases with λ and decreases with a.

6. DIFFRACTION DUE TO A CIRCULAR APERTURE (BRIEF)

Produces a central bright spot (Airy disc) surrounded by concentric dark and bright rings (Airy pattern).



Radius of first dark ring,

$$\sin\theta = 1.22\frac{\lambda}{D}$$

where D = diameter of aperture

7. KEY POINTS

- ★ Diffraction is significant when size ≈ λ.
- ★ Central maximum is brightest and widest.
- ★ Minima condition: $a\sin\theta = n\lambda$
- ★ Maxima condition: $a\sin\theta = \left(n + \frac{1}{2}\right)\lambda$
- ★ Diffraction pattern is symmetric about the central maximum.

★ Diffraction of light is a powerful demonstration of the wave nature of light.



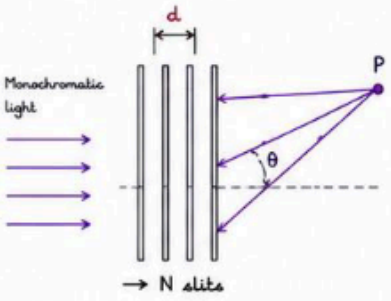
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DIFFRACTION GRATING & RESOLVING POWER

A diffraction grating is an optical device consisting of a large number of parallel, equally spaced slits. It produces sharp and intense maxima due to interference.

1. DIFFRACTION GRATING

- ★ A grating has a very large number of equally spaced, parallel slits.
- ★ Distance between two consecutive slits is called grating element (d).
- ★ It acts as a large number of coherent sources.
- ★ Produces principal maxima at certain angles.



d = grating element
N = total number of slits

2. GRATING EQUATION

$$d \sin\theta = n\lambda$$

where,
- d = grating element (spacing between slits)
- θ = angle of diffraction
- λ = wavelength of light
- n = order of maximum (0, ±1, ±2, ±3, ...)

Principal maxima (bright fringes):
- n = 0, ±1, ±2, ...
- n = 0 → central maximum (θ = 0°)
- n = ±1 → first order maxima
- n = ±2 → second order maxima
- and so on.

3. ANGULAR DISPERSION (D_θ)

It is the rate of change of angle with wavelength.

$$D_\theta = \frac{d\theta}{d\lambda} = \frac{n}{d\cos\theta}$$

Higher the order (n), greater is the angular dispersion.

4. DISPERSIVE POWER (D)

It is the ability of grating to separate two close wavelengths.

$$D = \frac{d\theta}{d\lambda} = \frac{n}{d\cos\theta}$$

⇒ Increases with order n
⇒ Decreases with d and cos θ

5. RESOLVING POWER (R)

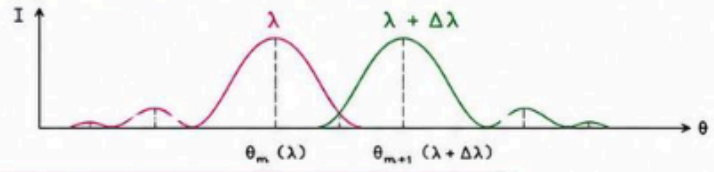
It is the ability of a grating to resolve (separate) two close wavelengths.

$$R = \frac{\lambda}{\Delta\lambda} = nN$$

where,
- λ = mean wavelength
- Δλ = least difference in wavelength that can be resolved
- N = total number of slits illuminated

6. RAYLEIGH CRITERION FOR RESOLUTION

Two wavelengths are just resolved when the principal maximum of one coincides with the first minimum of the other.



Condition for just resolution:

$$d(\sin\theta_{m+1} - \sin\theta_m) = \lambda$$ (Rayleigh's criterion)

Using grating equation,

$$R = \frac{\lambda}{\Delta\lambda} = nN$$

To increase resolving power:
- ★ Increase order (n)
- ★ Increase number of slits (N)

7. KEY POINTS

- ★ Grating produces sharp and intense principal maxima.
- ★ Condition for maxima: $d\sin\theta = n\lambda$
- ★ Central maximum: n = 0 at θ = 0°.
- ★ Angular dispersion: $D_\theta = n\frac{\lambda}{d\cos\theta}$
- ★ Dispersive power increases with n, decreases with d.
- ★ Resolving power: R = nN
- ★ Greater resolving power → better ability to separate close wavelengths.

★ Diffraction gratings are widely used in spectrometers to obtain high resolution spectra of light.



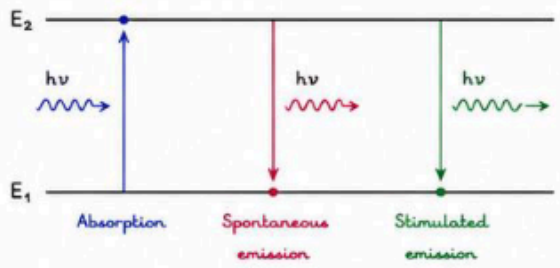
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LASERS – PRINCIPLES & EINSTEIN COEFFICIENTS

Laser stands for "Light Amplification by Stimulated Emission of Radiation".
It is a device that produces an intense, monochromatic, coherent and highly directional beam of light.

1. INTERACTION OF RADIATION WITH MATTER

When radiation of energy $h\nu$ interacts with an atom, three processes are possible.

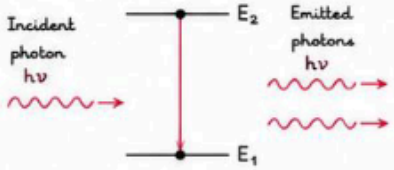


E_2 : Higher energy level
E_1 : Lower energy level
h : Planck's constant
ν : Frequency of radiation

★ Stimulated emission is the key process behind laser action.

2. STIMULATED EMISSION

An incident photon of energy $h\nu$ (where $h\nu = E_2 - E_1$) falling on an atom in excited state (E_2) can stimulate it to come to lower level (E_1), emitting a photon of the same energy, phase, direction and polarization.



★ The two photons are identical.
★ Leads to amplification of light.
★ Basis of laser action.

3. POPULATION INVERSION (NECESSARY CONDITION)

Under normal conditions, more atoms are in lower energy level.
For laser action, we need more atoms in excited state.

$N_2 > N_1$

N_2 : Number of atoms in E_2
N_1 : Number of atoms in E_1

4. METASTABLE STATE

★ Lasers use an intermediate metastable energy level.
★ Atoms are first excited to this level, which has a long lifetime ($\sim 10^{-3}$ to 10^{-8} s).
★ This allows population inversion to be achieved.

5. BASIC PRINCIPLE OF LASER

Energy Source (Excitation) → Active Medium (Gain Medium) → Optical Resonator (Cavity) → Coherent Light (Laser Beam)

★ Active medium provides amplification by stimulated emission.
★ Optical resonator provides feedback and selects modes.
★ A fraction of light escapes as laser output.

6. EINSTEIN COEFFICIENTS

Einstein (1917) introduced three coefficients to describe interaction of radiation with matter.

Process	Einstein Coefficient	Meaning	Rate (per atom)
Absorption	B_{12}	$E_1 \rightarrow E_2$	$B_{12}\,\rho(\nu)$
Spontaneous Emission	A_{21}	$E_2 \rightarrow E_1$	A_{21}
Stimulated Emission	B_{21}	$E_2 \rightarrow E_1$	$B_{21}\,\rho(\nu)$

$\rho(\nu)$: Energy density of radiation at frequency ν.

7. RELATION BETWEEN EINSTEIN COEFFICIENTS

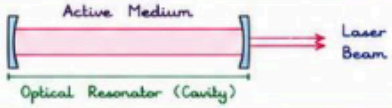
$$\frac{g_1 B_{12}}{g_2 B_{21}} = 1 \quad \text{and} \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$$

g_1, g_2 : Statistical weights (degeneracies) of E_1 and E_2
c : Speed of light

8. CHARACTERISTICS OF LASER LIGHT

★ Highly monochromatic (very small $\Delta\lambda$)
★ Highly coherent (spatial and temporal)
★ Highly directional (very low divergence)
★ Very high intensity (high brightness)
★ Can be focused to a very small spot

★ Laser action is based on stimulated emission and population inversion, enabled by Einstein coefficients and optical feedback.

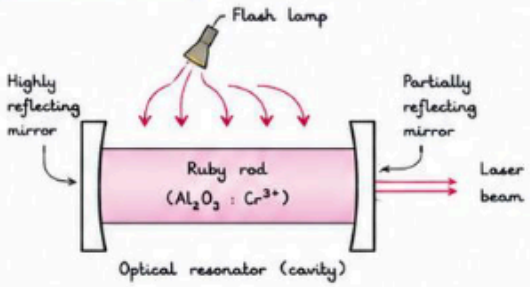


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LASER TYPES & APPLICATIONS

Lasers are classified based on the active medium used. Different types of lasers have different wavelengths and are used for various engineering and scientific applications.

1. RUBY LASER



Principle / Working

- ★ Cr^{3+} ions in ruby rod are pumped to excited state by intense light from flash lamp.
- ★ Population inversion is achieved.
- ★ Stimulated emission produces coherent red light.

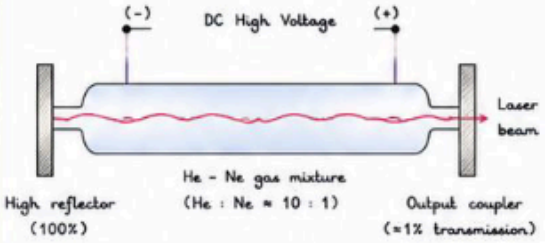
Key Features

- Wavelength : 694.3 nm (Red light)
- Pulsed laser
- High peak power
- First laser ever developed (1960 by Maiman)

Applications

- ★ Material processing (drilling, cutting, welding)
- ★ Holography
- ★ Medical surgery (e.g., eye surgery)
- ★ Scientific research

2. He - Ne LASER



Principle / Working

- ★ A low pressure mixture of He and Ne gas is excited by electric discharge.
- ★ Energy transfer from He to Ne excites Ne atoms.
- ★ Population inversion in Ne levels produces stimulated emission.

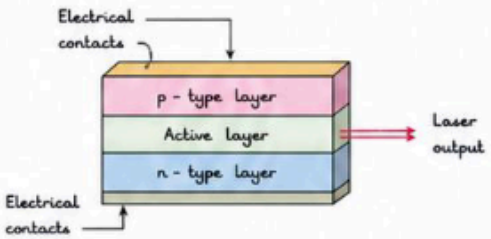
Key Features

- Wavelength : 632.8 nm (Red light)
- Continuous wave (CW) laser
- Very stable and narrow linewidth
- Low power output

Applications

- ★ Interferometry
- ★ Alignment in labs
- ★ Barcode scanners
- ★ Metrology
- ★ Holography

3. SEMICONDUCTOR LASER (DIODE LASER)



Principle / Working

- ★ p-n junction is forward biased.
- ★ Electrons from n-side and holes from p-side recombine in active region.
- ★ Stimulated emission occurs due to population inversion.

Key Features

- Wavelength : 0.8 - 1.6 μm (IR range)
- Compact and efficient
- Low power consumption
- Easy to modulate
- Long life and reliable

Applications

- ★ Fiber optic communication
- ★ CD / DVD players
- ★ Laser printers
- ★ Barcode scanners
- ★ Medical instruments

4. GENERAL APPLICATIONS OF LASERS

- ★ Industrial : Cutting, welding, drilling, engraving, surface hardening
- ★ Medical : Surgery, dermatology, dental, eye treatment (LASIK)
- ★ Communication : Optical fiber communication
- ★ Research : Spectroscopy, holography, interferometry
- ★ Defense : Target designation, range finding, missile guidance
- ★ Entertainment : Laser shows, projection systems
- ★ Data storage : CD, DVD, Blu-ray
- ★ Scientific instruments : LIDAR, laser microscopes

★ Different laser types are designed for specific wavelengths and output powers to suit specific applications in science, engineering and technology.

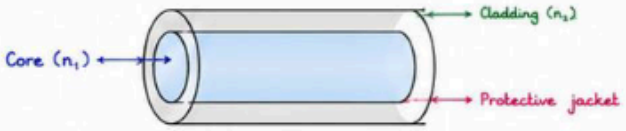


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OPTICAL FIBER – PRINCIPLES & NUMERICAL APERTURE

An optical fiber is a thin, flexible, transparent medium that guides light from one point to another using the principle of total internal reflection (TIR).

1. BASIC STRUCTURE OF OPTICAL FIBER

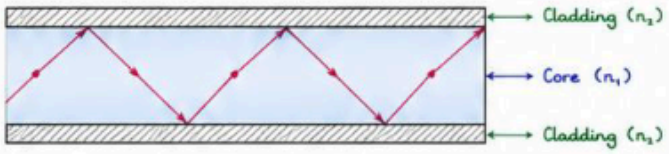


- ★ Core : Central region through which light travels.
- ★ Cladding : Surrounds the core with a slightly lower refractive index.
- ★ Protective jacket : Provides mechanical strength and protection.

Refractive index of core (n_1) > refractive index of cladding (n_2)

2. TOTAL INTERNAL REFLECTION (TIR)

Light is guided through the core by repeated total internal reflections at the core-cladding boundary.



Condition for TIR : $\theta_i > \theta_c$

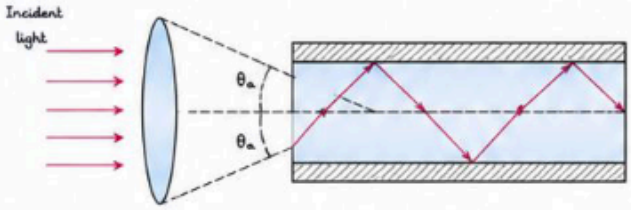
where,

θ_i = angle of incidence at core-cladding boundary

θ_c = critical angle

$$\sin\theta_c = \frac{n_2}{n_1} \quad (n_1 > n_2)$$

3. ACCEPTANCE ANGLE & ACCEPTANCE CONE



- ★ Maximum angle at which light can enter the fiber is called acceptance angle (θ_a).
- ★ The cone formed by this angle is called acceptance cone.

4. NUMERICAL APERTURE (N.A.)

The ability of a fiber to accept light is measured by its numerical aperture.

$$N.A. = n_0 \sin\theta_a = \sqrt{n_1^2 - n_2^2}$$

where,

n_0 = refractive index of the medium outside the fiber (usually air, $n_0 \approx 1$)

n_1 = refractive index of core

n_2 = refractive index of cladding

θ_a = acceptance angle in air

Higher N.A. → more light gathering ability of the fiber.

5. RELATION BETWEEN PARAMETERS

$$\sin\theta_c = \frac{n_2}{n_1} \Rightarrow \sin\theta_a = \frac{\sqrt{n_1^2 - n_2^2}}{n_0} \Rightarrow N.A. = n_0 \sin\theta_a$$

- ★ For air ($n_0 = 1$): $N.A. = \sin\theta_a = \sqrt{n_1^2 - n_2^2}$
- ★ Condition for guiding light: $n_1 > n_2$

6. KEY POINTS

- ★ Optical fiber works on the principle of total internal reflection.
- ★ Core has higher refractive index than cladding ($n_1 > n_2$).
- ★ Light enters the core within acceptance cone (θ_a).
- ★ Numerical aperture (N.A.) decides the light gathering capacity.
- ★ Greater the difference between n_1 and n_2, greater will be the N.A.

7. IMPORTANT FORMULAS

Critical angle : $\sin\theta_c = \frac{n_2}{n_1}$ $(n_1 > n_2)$ | Acceptance angle : $\sin\theta_a = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$ | Numerical aperture : $N.A. = n_0 \sin\theta_a = \sqrt{n_1^2 - n_2^2}$

★ Optical fibers guide light efficiently using total internal reflection and are the backbone of modern high-speed communication systems.



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OPTICAL FIBER – TYPES, LOSSES & DISPERSION

In optical fiber communication, signal strength reduces due to losses and pulse spreads due to dispersion. Both limit the performance of a fiber link.

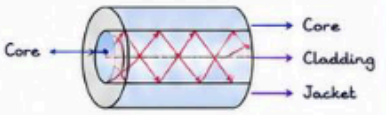
1. TYPES OF OPTICAL FIBER

(A) Single Mode Fiber (SMF)



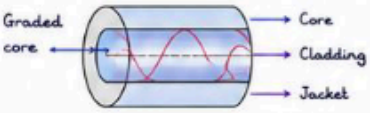
- ★ Core diameter : 8 – 10 μm
- ★ Cladding diameter : 125 μm
- ★ Carries only one mode
- ★ Used for long distance (low loss, high bandwidth)

(B) Multimode Step-Index Fiber (MM-SI)



- ★ Core diameter : 50 or 62.5 μm
- ★ Cladding diameter : 125 μm
- ★ Many modes travel by total internal reflection
- ★ Used for short distance (higher modal dispersion)

(C) Multimode Graded-Index Fiber (MM-GI)



- ★ Core diameter : 50 or 62.5 μm
- ★ Refractive index decreases gradually from center to cladding
- ★ Modes follow curved paths
- ★ Less modal dispersion than step-index

2. LOSSES IN OPTICAL FIBER

Type of Loss	Cause	Dependence / Notes
Absorption Loss	Material absorbs light (impurities, OH^- ions, metals)	Increases with wavelength (more in UV & IR) Minimum around 1.3 μm and 1.55 μm
Scattering Loss	Due to microscopic inhomogeneities in material	Rayleigh scattering $\propto \frac{1}{\lambda^4}$
Bending Loss	Bends cause some rays to escape from the core	(a) Macrobending (large scale bends) (b) Microbending (small irregular bends)
Radiation Loss	Imperfections at joints, splices, connectors	Light leaks out
Coupling Loss	Mismatching between source-fiber or fiber-fiber	Due to difference in N.A., core size, alignment

★ Total attenuation : $\alpha_{total} = \alpha_{abs} + \alpha_{scat} + \alpha_{bend} + \alpha_{rad} + \alpha_{coup}$ (dB/km)

Power decreases as : $P(z) = P(0)\, e^{-\alpha z}$

3. DISPERSION IN OPTICAL FIBER

Dispersion causes pulse broadening → limits bandwidth.

Total Dispersion = Modal Dispersion + Chromatic Dispersion

(A) Modal Dispersion (in Multimode Fibers)

★ Different modes take different path lengths → arrive at different times.

Pulse broadening : $\Delta t_m = \frac{n_1}{c}(NA)^2$

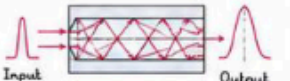
where,

n_1 = refractive index of core

Δ = relative index difference = $\frac{(n_1 - n_2)}{n_1}$

NA = numerical aperture

c = speed of light



(B) Chromatic Dispersion (in All Fibers)

★ Different wavelengths travel at different speeds due to material dispersion and waveguide dispersion.

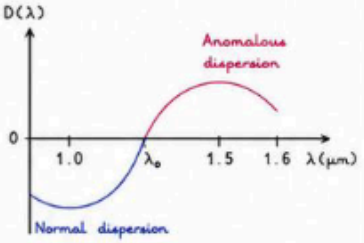
Pulse broadening : $\Delta t_c = D(\lambda)\, L\, \Delta\lambda$

where,

$D(\lambda)$ = dispersion parameter (ps / (nm·km))

L = length of fiber (km)

$\Delta\lambda$ = spectral width of source (nm)



$\lambda_0 \approx 1.3$ μm for silica fiber (Zero dispersion wavelength)

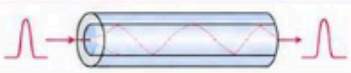
4. EFFECTS OF DISPERSION

- ★ Pulse broadening → overlap between consecutive pulses (Inter-symbol interference).
- ★ Limits the data rate × distance product (Bandwidth-length product).
- ★ More severe in multimode fibers.

5. REDUCTION OF LOSSES & DISPERSION

To Reduce Losses	To Reduce Dispersion
★ Use high purity silica.	★ Use single mode fiber for long distance.
★ Operate at 1.3 μm or 1.55 μm.	★ Use graded-index fiber for multimode.
★ Proper bending radius.	★ Use narrow spectral width sources (LED with small $\Delta\lambda$, Laser).
★ Good quality connectors & splices.	
★ Match source and fiber parameters.	

★ Low loss and low dispersion enable high data rate and long distance communication.



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FORMULA SHEET – PART 1

A quick formula sheet for easy revision of important concepts in Wave Optics, Diffraction, Lasers and Optical fibers.

1. INTERFERENCE

★ Path difference

$$\Delta = d \sin\theta$$

★ Constructive (Bright)

$$\Delta = n\lambda \quad (n = 0, \pm1, \pm2, \ldots)$$

★ Destructive (Dark)

$$\Delta = (n + \tfrac{1}{2})\lambda$$

★ Intensity

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$$

2. THIN FILMS

★ Condition for maxima (Reflected)

$$2\mu t \cos r = (2n+1)\frac{\lambda}{2}$$

★ Condition for minima (Reflected)

$$2\mu t \cos r = n\lambda$$

μ = refractive index of film
t = thickness of film
r = angle of refraction in film

3. DIFFRACTION (SINGLE SLIT)

★ Minima (Dark fringes)

$$a \sin\theta = n\lambda \quad (n = \pm1, \pm2, \ldots)$$

★ Angular width of central maximum

$$\theta_0 = \frac{2\lambda}{a} \quad \text{(radians)}$$

a = width of slit
λ = wavelength

4. DIFFRACTION GRATING

★ Grating Equation

$$d \sin\theta = n\lambda$$

where,

d = grating element (spacing)
θ = diffraction angle
n = order of maximum $(0, \pm1, \pm2, \ldots)$

5. DISPERSESION & RESOLVING POWER

★ Angular Dispersion

$$D_\theta = \frac{d\theta}{d\lambda} = \frac{n}{d\cos\theta}$$

★ Dispersive Power

$$D = \frac{d\theta}{d\lambda} = \frac{n}{d\cos\theta}$$

★ Resolving Power

$$R = \frac{\lambda}{\Delta\lambda} = nN$$

N = total number of slits illuminated

6. LASERS (EINSTEIN COEFFICIENTS)

Process	Einstein Coefficient	Transition	Rate (per atom)
Absorption	B_{12}	$E_1 \to E_2$	$B_{12}\,\rho(\nu)$
Spontaneous Emission	A_{21}	$E_2 \to E_1$	A_{21}
Stimulated Emission	B_{21}	$E_2 \to E_1$	$B_{21}\,\rho(\nu)$

Relation between coefficients :

$$\frac{g_1 B_{12}}{g_2 B_{21}} = 1 \quad \text{and} \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$$

g_1, g_2 = statistical weights of lower & upper levels
$\rho(\nu)$ = energy density of radiation at frequency ν

7. OPTICAL FIBER – NUMERICAL APERTURE

★ Numerical Aperture (N.A.)

$$N.A. = n_0 \sin\theta_a = \sqrt{n_1^2 - n_2^2}$$

where,

n_0 = refractive index of medium outside fiber
n_1 = refractive index of core
n_2 = refractive index of cladding
θ_a = acceptance angle in air

8. OPTICAL FIBER – TIR & PARAMETERS

★ Condition for Total Internal Reflection

$$\theta_i > \theta_c$$

where $\sin\theta_c = \frac{n_2}{n_1}$ $\quad (n_1 > n_2)$

★ Acceptance Angle

$$\sin\theta_a = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

★ V-number (Normalized Frequency)

$$V = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2}$$

a = core radius
λ = wavelength

9. DISPERSION IN FIBERS

★ Total Dispersion

$$D_T = D_M + D_C$$

D_M = material (chromatic) dispersion
D_C = waveguide dispersion

★ Pulse Broadening

$$\Delta t = D(\lambda)\, L\, \Delta\lambda$$

$D(\lambda)$ = dispersion parameter
L = fiber length
$\Delta\lambda$ = spectral width of source

QUICK REVISION – KEY POINTS

WAVE OPTICS
- Interference due to superposition of waves.
- Path diff.: $\Delta = d\sin\theta$
- Bright: $\Delta = n\lambda$
- Dark: $\Delta = (n+\frac{1}{2})\lambda$
- Intensity depends on phase difference.

THIN FILMS
- Interference in thin films due to reflection.
- Maxima (R): $2\mu t\cos r = (2n+1)\frac{\lambda}{2}$
- Minima (R): $2\mu t\cos r = n\lambda$

DIFFRACTION
- Single slit minima: $a\sin\theta = n\lambda$
- Central max. width: $\theta_0 = \frac{2\lambda}{a}$
- Grating eqn.: $d\sin\theta = n\lambda$
- Resolving power: $R = nN$

LASERS
- Based on stimulated emission.
- Population inversion is essential.
- Einstein coefficients A_{21}, B_{12}, B_{21} describe interactions.

OPTICAL FIBER
- Guides light by TIR.
- N.A. = $\sqrt{n_1^2 - n_2^2}$
- Higher N.A. → more light gathering.
- Low losses & dispersion → high performance.

LOSSES & DISPERSION
- Losses: absorption, scattering, bending, radiation, coupling.
- Dispersion: modal + chromatic.
- Limit bandwidth & cause pulse broadening.

★ Master the concepts, practice numericals and draw diagrams for better understanding!

